

Shane Sheth

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Education

University of Southern California, B.S. Mechanical Engineering, Dean's List

Expected Graduation: May 2029

GPA: 3.63

Technical Skills

Software (analysis/CAD) - SolidWorks, Onshape, MATLAB, Python, Google Workspace (Sheets, Docs, Slides)

Manufacturing - Composite Layups, CNC Laser Cutter, CNC Water Jet, Hot Wire, Soldering, Welding, Plasma Cutter

Experience

Configurations Lead - USC AeroDesign Team (AIAA Design/Build/Fly)

May 2026 - Present

- Lead high-level aircraft configuration downselects for a 25+ member design team, determining aircraft architecture, wing and payload positioning, and propulsion, tail, and fuselage configuration
- Design full-aircraft CAD model in SolidWorks, balancing aerodynamic, structural, performance, landing-gear, and payload constraints, and determining tradeoffs to blend competing subsystem constraints into a single manufacturable aircraft and translating finalized geometry into build drawings, laser-cut documents, and molds for manufacturing
- Redesigned aircraft tail after manufacturing exposed structural issues, reinforcing under-supported regions while removing unnecessary structure from overbuilt components to improve assembly and reduce tail weight by 8 g (7%)
- Optimized control-surface rib-to-spar joints after assembly exposed tolerance and stiffness issues, reversing the interlocking geometry to improve alignment and structural engagement

Structures Engineer - USC AeroDesign Team (AIAA Design/Build/Fly)

Aug 2025 - May 2026

- Redesigned the empennage from 3 separate components into a singular structure mounted to the tail boom at 2 locations, simplifying aircraft integration and improving load transfer through fiberglass-plywood sandwich hardpoints and by extending vertical stabilizer spars into the horizontal stabilizer root rib. Incorporated a 4° horizontal stabilizer AoI and tail fairing to minimize interference drag. Reduced empennage weight by 9.5% on aircraft highly sensitive to empty weight by designing structurally determined weight cutouts, minimizing integration hardware, and replacing CF D-box skin with lighter material
- Led 10 validation tests for a passenger restraint system on a 1:1 testbed with a FoS of 1.5, by deriving an 8G maximum load factor from flight test data and comparing multiple restraint designs to recommend the final configuration, verifying zero measurable payload displacement required for aircraft acceptance during competition technical inspection
- Co-authored 2 sections in the 1st place 2026 AIAA Design/Build/Fly competition report detailing payload design decisions, testing methodologies, and experimental results

Electrical Systems Co-Lead - Terra Labs

Feb 2026 - Present

- Co-led electrical development across a team of 9 engineers for a 7-DOF autonomous robotic arm (Cove), establishing design requirements, schematic/wiring diagrams, and test procedures, to ensure reliable communication between Teensy 4.1, Raspberry Pi 5, and CAN bus actuator network
- Designed and led testing for Cove's 48 V, 480 W power distribution system, sizing battery capacity, wire gauges, XT60/XT30 connectors, and per-actuator fusing from peak current, voltage-drop, and safety-factor calculations, enabling reliable power delivery to actuators while minimizing wire size to preserve arm flexibility
- Diagnosed power-distribution issues during system integration and redesigned the actuator supply into 3 individually fused parallel branches, improving load isolation and preventing a single branch from overloading shared wiring or disrupting power to the remaining actuators
- Designed and debugged Cove's hardware E-stop system as one layer of a dual shutdown system, integrating a normally closed emergency-stop path with the system's software shutdown layer and validating that operators could physically interrupt actuator power during testing

Global Summit Intern - Northrop Grumman

June 2024 - July 2024

- Modified a 59-in-span twin-motor radio-controlled aircraft for a firefighting mission as part of a 10-member team, designing payload mounting and aircraft interfaces in Onshape to integrate the fire suppressant payload delivery system
- Analyzed results from 4+ flight tests to identify weight, landing-gear, and payload-integration issues, removing nonstructural foam and the nose-wheel servo to target a 5.5-lb aircraft mass and 0.921 g/cm² wing loading versus a projected 7.0-lb, 1.172 g/cm² configuration
- Co-authored and presented aircraft integration, flight test findings, and design changes at a Critical Design Review to Northrop Grumman VPs

Projects

Experience Synesthesia - Artscape (USC Performing Arts Committee)

Nov 2025 - March 2026

- Designed a 26-ft interactive projection structure from fourteen 3/8-in thick pine plywood panels, developing a modular double-arc frame to withstand 12-mph wind gusts while allowing repeated transport and <24-hour installation with limited manpower
- Designed three interactive podiums and produced manufacturing drawings for fabrication, each supporting 6.16 kg of integrated hardware at a >3.0 factor of safety while accommodating users of varying heights

Band Together - Artscape (USC Performing Arts Committee)

Nov 2025 - March 2026

- Converted an artistic concept into the structural design for an 18 × 18-ft interactive art installation for 3,000+ festival attendees, defining structural design, material selection, manufacturing methods, and a guitar-shaped layout for intuitive audience interaction
- Developed control architecture integrating 10 photoelectric sensors, 5 instrument stations, 3 speakers, and a Raspberry Pi to trigger synchronized audio and lighting as participants moved through the installation, enabling users to progressively layer a musical composition

Hard Top Surfboards

Aug 2024 - June 2025

- Shaped two custom EPS-base surfboards (9'2" x 21.5", 5'8" x 19"). Designed custom planform, rocker, rail, and bottom-contour geometry
- Fabricated fiberglass/epoxy composite skins using a 1-ply bottom and 2-ply deck ply schedule. Refined resin application and rail wrapping between builds to reduce air pockets, excess epoxy, and surface irregularities